

Solutions to Problem Set 5

1. In a single-parity-check code, a single parity bit b_1 is appended to a block of k information bits, $m_1, m_2, \dots, m_k$. The parity bit is chosen so that the codeword satisfies the even parity rule:

$$m_1 + m_2 + \dots + m_k + b_1 = 0, \quad \text{mod } 2$$

- a) For $k = 3$, construct the 2^k valid codewords in the code defined by this rule.
b) Show that the minimum Hamming distance among valid codewords is $d_{\min} = 2$.

Solution. a) The valid codewords are 0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111.

- b) We prove by contradiction. Suppose there are two valid codewords $\{m_1^{(1)}, \dots, m_k^{(1)}, b_1^{(1)}\}$ and $\{m_1^{(2)}, \dots, m_k^{(2)}, b_1^{(2)}\}$ with Hamming distance 1. There is exactly one different bit, i.e.,

$$(m_1^{(1)} + m_1^{(2)}) + \dots + (m_k^{(1)} + m_k^{(2)}) + (b_1^{(1)} + b_1^{(2)}) = 1, \quad \text{mod } 2$$

However, each codeword satisfying the even parity rule implies that the sum satisfying the rule, i.e.,

$$(m_1^{(1)} + m_2^{(1)} + \dots + m_k^{(1)} + b_1^{(1)}) + (m_1^{(2)} + m_2^{(2)} + \dots + m_k^{(2)} + b_1^{(2)}) = 0, \quad \text{mod } 2$$

Contradiction! Therefore, the minimum distance is 2.

2. Consider a linear block code with total length $n = 13$ bits. If the code tolerates up to 3 bit errors, at least how many parity check bits is needed?

Solution. Since the code tolerates up to 3 bit errors, the total number of possible error patterns to identify are

$$C_{13}^1 + C_{13}^2 + C_{13}^3 = 377$$

Then the number of parity bits should satisfy

$$2^r - 1 \geq 377, \quad r \geq 9$$

3. Consider a (7,4) Hamming code with generator matrix

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

a) Determine the parity check matrix $\mathbf{H}$.

b) Decode the message if the received codeword is 0101011.

Solution. a) Since $\mathbf{G} = [\mathbf{I}|\mathbf{Q}]$,

$$\mathbf{H} = [\mathbf{Q}^T|\mathbf{I}] = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

b) The syndrome is

$$\mathbf{s} = \mathbf{b}\mathbf{H}^T = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}.$$

Thus the received codeword is correct and the message is the first 4 bits, 0101.

4. Consider a (7,3) linear block code with generator matrix

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- Construct the generator matrix $\bar{\mathbf{G}}$ of an equivalent systematic code and brief on why they are equivalent. (Hint: Conduct row transformations on $\mathbf{G}$)
- Determine the minimum Hamming distance $d_{\min}$. What is the error correction capability t and the error detection capability e ?
- Can you decode the message if the received codeword is $r_1 = 0100010$ or $r_2 = 1111001$?

Solution. a) If we add an identity matrix to $\mathbf{G}$ and conduct row transformations,

$$\begin{aligned} [\mathbf{G}|\mathbf{I}] &= \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & | & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 1 & 0 & | & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & | & 1 & 0 & 0 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 & | & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & | & 1 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 & | & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & | & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & | & 1 & 0 & 0 \end{bmatrix} = [\bar{\mathbf{G}}|\mathbf{R}] \end{aligned}$$

where $\bar{\mathbf{G}}$ is the transformed generator matrix of the systematic code, and $\bar{\mathbf{G}} = \mathbf{R}\mathbf{G}$.

Equivalently, we can use $\bar{\mathbf{G}}$ to analyze the code since $\mathbf{x}\mathbf{G} = (\mathbf{x}\mathbf{R}^{-1})\bar{\mathbf{G}}$. Therefore, the whole coding pipeline is: first multiply $\mathbf{x}$ by $\mathbf{R}^{-1}$, then encode using $\bar{\mathbf{G}}$, after the noisy channel, decode using $\bar{\mathbf{H}}$ and finally multiply by $\mathbf{R}$ to recover the original message.

- By trial, the minimum Hamming distance is $d_{\min} = 4$. Thus $2t + 1 \leq 4$ implies the error correction capability $t = 1$, and $t + e + 1 \leq 4$ implies the error detection capability $e = 2$. Also note that the code can detect $e' = 3$ errors without any error correction capability.

- Let $\bar{\mathbf{G}} = [\mathbf{I}|\mathbf{Q}]$, then

$$\bar{\mathbf{H}} = [\mathbf{Q}^T|\mathbf{I}] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

If the received codeword is $\mathbf{r}_1 = 0100010$,

$$\mathbf{s}_1 = \mathbf{r}_1\bar{\mathbf{H}}^T = \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix}.$$

Since none of the columns of $\bar{\mathbf{H}}$ corresponds to $\mathbf{s}_1$, it is not decodable, although we are sure there is at least two bit errors.

If the received codeword is $\mathbf{r}_2 = 1111001$,

$$\mathbf{s}_2 = \mathbf{r}_2 \bar{\mathbf{H}}^T = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}.$$

Since $\mathbf{s}_2$ corresponds to the first row of $\bar{\mathbf{H}}$, the error happens at the first bit. Therefore the decoded message is 011. Finally, to recover the original message, we multiply by $\mathbf{R}$ to obtain 010.